

Experiment No-6

Basic Operations on Signals and sequences

AIM: perform the operations on signals and sequences such as addition, multiplication, scaling, shifting, folding and also compute energy and power.

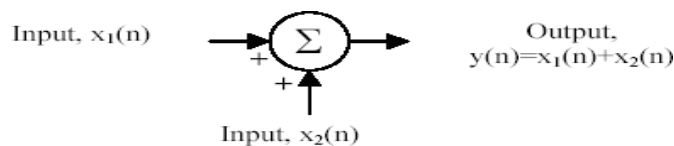
Software Required: Matlab software.

Theory:

Signal Addition

Addition: any two signals can be added to form a third signal, z

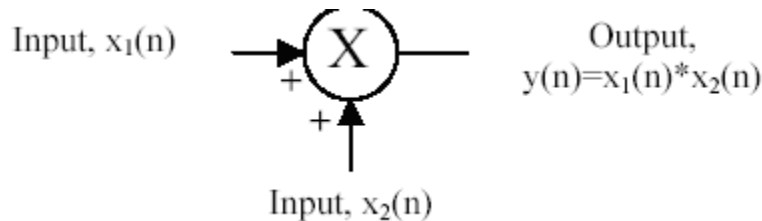
$$z(t) = x(t) + y(t)$$



Multiplication :

Multiplication of two signals can be obtained by multiplying their values at every instants . z

$$z(t) = x(t) y(t)$$



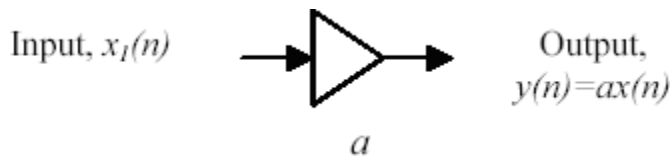
Time reversal/Folding:

Time reversal of a signal $x(t)$ can be obtained by folding the signal about $t=0$.

$$Y(t) = x(-t)$$

Signal Amplification/Scaling : $Y(n) = ax(n)$ if $a < 1$ attnuation

$a > 1$ amplification



Time shifting: The time shifting of $x(n)$ obtained by delay or advance the signal in time by using $y(n) = x(n+k)$

If k is a positive number, y(n) shifted to the right i.e. the shifting delays the signal

If k is a negative number, y(n) it gets shifted left. Signal Shifting advances the signal

Times New RomanEnergy:

$$E[x] = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Average power:

$$P[x] = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Program:

```
clc;

clear all;close
all;
%~~~~~

% generating two input signalst=0:.01:1;
x1=sin(2*pi*4*t);
x2=sin(2*pi*8*t);
subplot(2,2,1);
plot(t,x1);
xlabel('time');
ylabel('amplitude');
title('input signal 1');
subplot(2,2,2);
plot(t,x2);
xlabel('time');
ylabel('amplitude');
title('input signal 2');

% addition of signalsy1=x1+x2;
subplot(2,2,3);
plot(t,y1);
xlabel('time');
ylabel('amplitude');
title('addition of two signals');

% multiplication of signals
y2=x1.*x2;
```

```
subplot(2,2,4);  
plot(t,y2);  
xlabel('time');  
ylabel('amplitude');  
title('multiplication of two signals');
```

```
% scaling of a signal1
```

```
A=2;  
y3=A*x1;  
figure;  
subplot(2,2,1);  
plot(t,x1);  
xlabel('time');  
ylabel('amplitude');  
title('input signal')  
subplot(2,2,2);  
plot(t,y3);  
xlabel('time');  
ylabel('amplitude');  
title('amplified input signal');
```

```
% folding of a signal1
```

```
h=length(x1);  
nx=0:h-1;  
subplot(2,2,3);  
plot(nx,x1);  
xlabel('nx'); ylabel('amplitude');  
title('input signal')  
y4=fliplr(x1);  
nf=-fliplr(nx);  
subplot(2,2,4);  
plot(nf,y4);  
xlabel('nf');  
ylabel('amplitude');  
title('folded signal');
```

```
%shifting of a signal 1
```

```
figure; subplot(3,1,1);  
plot(t,x1);  
xlabel('time t');  
ylabel('amplitude');  
title('input signal');subplot(3,1,2);
```

```

plot(t+2,x1);
xlabel('t+2');
ylabel('amplitude');
title('right shifted signal');
subplot(3,1,3);
plot(t-2,x1);
xlabel('t-2');
ylabel('amplitude');
title('left shifted signal');
%~~~~~
%operations on sequences
n1=1:1:9;
s1=[1 2 3 0 5 8 0 2 4];
figure;
subplot(2,2,1);
stem(n1,s1);
xlabel('n1');
ylabel('amplitude');
title('input sequence1');
s2=[1 1 2 4 6 0 5 3 6];
subplot(2,2,2);
stem(n1,s2);
xlabel('n2'); ylabel('amplitude');
title('input sequence2');

% addition of sequences
s3=s1+s2; subplot(2,2,3);
stem(n1,s3);
xlabel('n1');
ylabel('amplitude');
title('sum of two sequences');

% multiplication of sequences
s4=s1.*s2;
subplot(2,2,4);
stem(n1,s4);
xlabel('n1');
ylabel('amplitude');
title('product of two sequences');
%~~~~~

```

```

% program for energy of a sequence
z1=input('enter the input sequence');
e1=sum(abs(z1).^2);
disp('energy of given sequence is');e1

% program for energy of a signal
t=0:pi:10*pi;
z2=cos(2*pi*50*t).^2;
e2=sum(abs(z2).^2);
disp('energy of given signal is');e2

% program for power of a sequence
p1= (sum(abs(z1).^2))/length(z1);
disp('power of given sequence is');p1

% program for power of a signal
p2=(sum(abs(z2).^2))/length(z2);
disp('power of given signal is');p2

```

OUTPUT:

enter the input sequence[1 3 2 4 1]

energy of given sequence is

e1 = 31

energy of given signal is

e2 = 4.0388

power of given sequence is

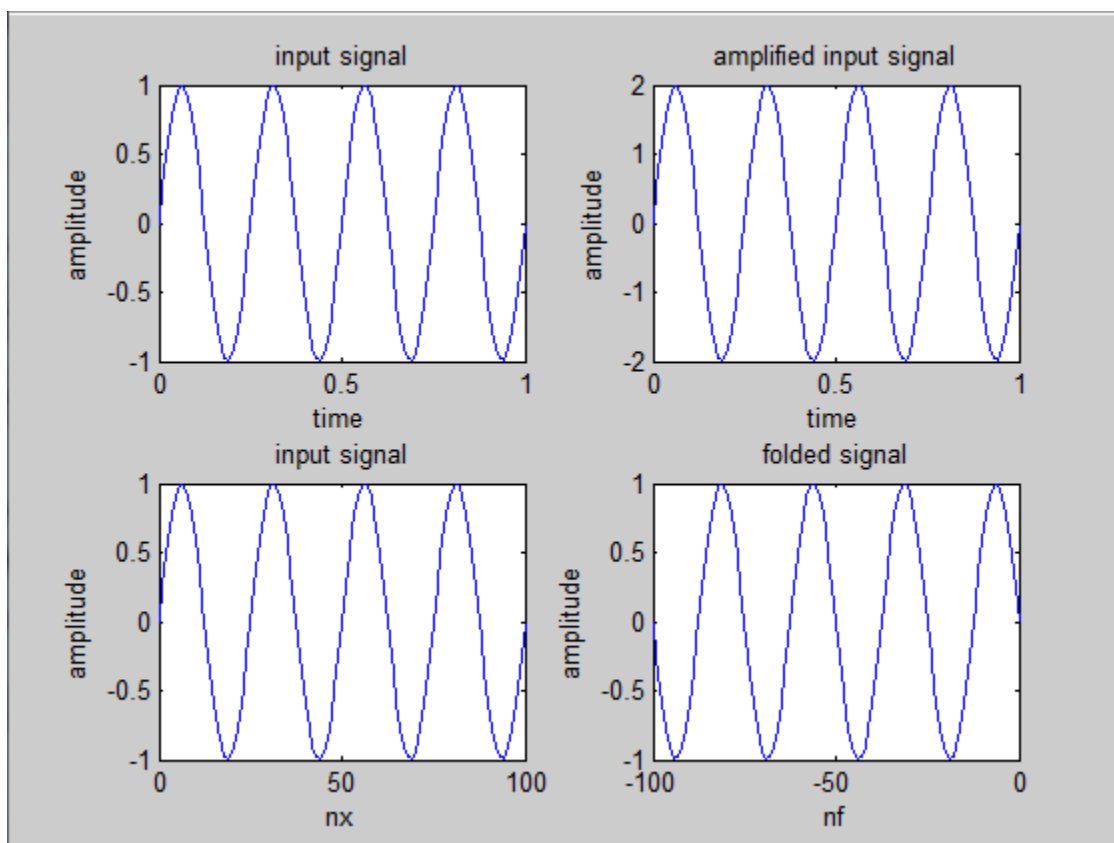
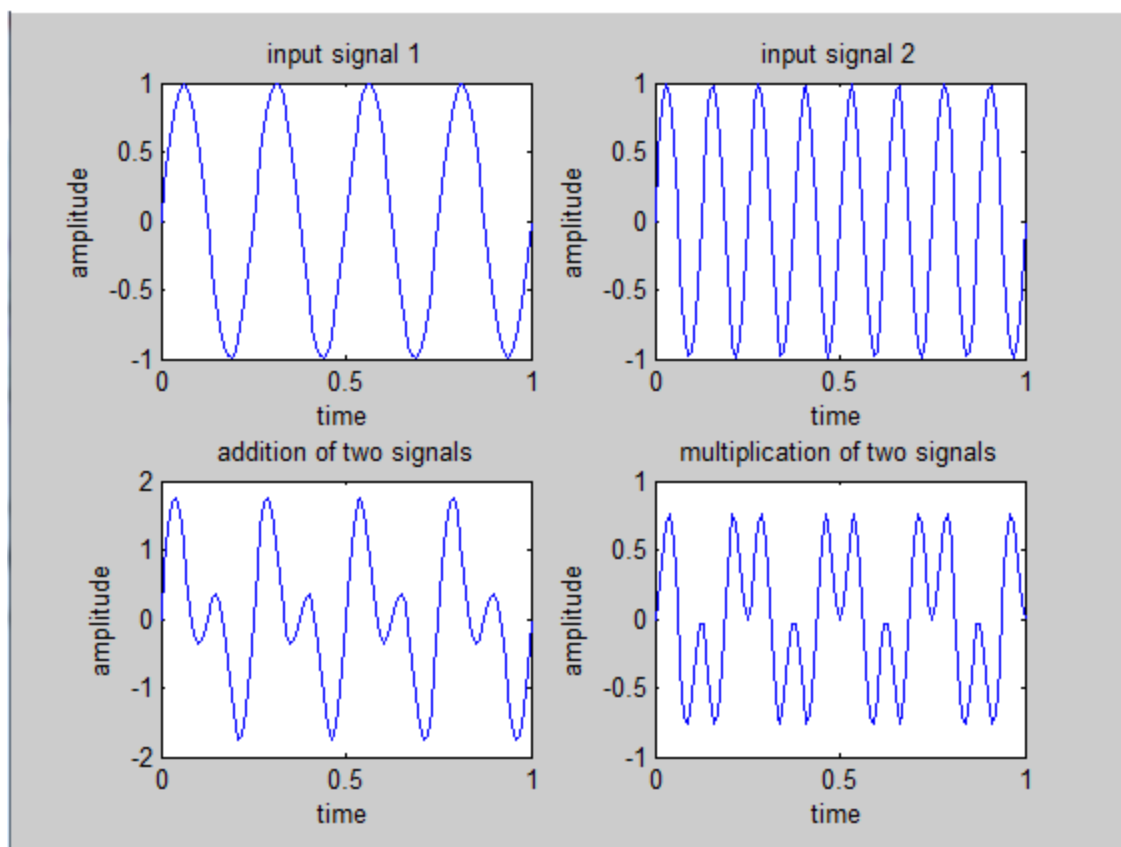
p1 = 6.2000

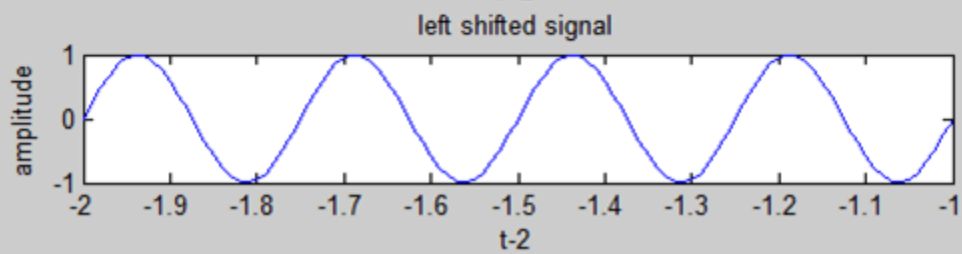
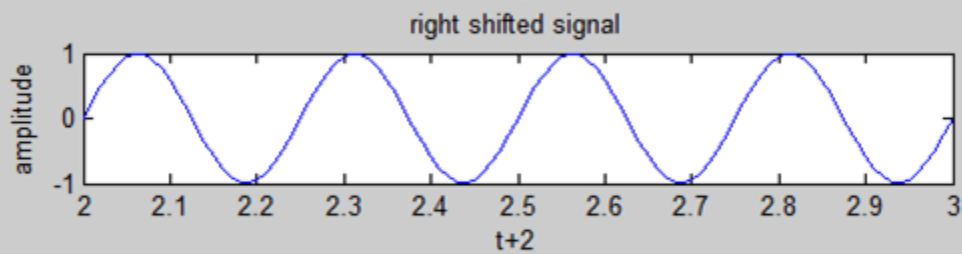
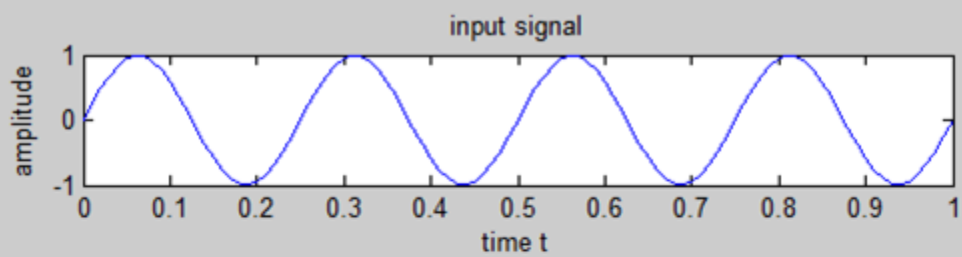
power of given signal is

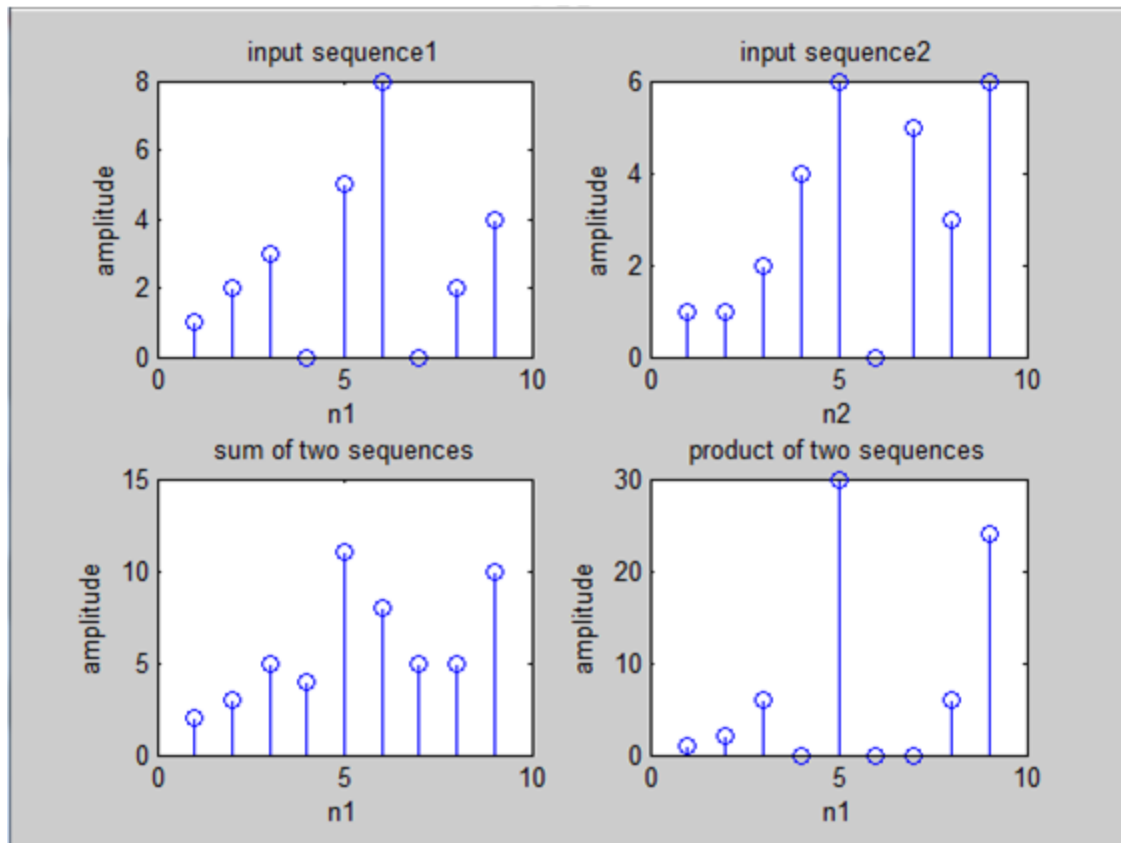
p2 = 0.3672

Result: Various operations on signals and sequences are performed.

Output:







VIVA QUESTIONS:-

1. Define Symmetric and Anti-Symmetric Signals?
2. Define Continuous and Discrete Time Signals?
3. What are the Different types of representation of discrete time signals?
4. What are the Different types of Operation performed on signals?

What is System?